Nilesh Gupta

Member of Technical Staff
Mirendil

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Google Scholar

Research Interests

My research focuses on End-to-end Information Retrieval and Efficient LLM architectures for very long-context modeling

Education

University of Texas at Austin *2021 - 2026*
Ph.D. in Computer Science and Engineering
Thesis: *Towards Unified and Scalable End-to-end Search in Large Output Spaces*
Advisor: Prof. Inderjit Dhillon
Committee: Prof. Greg Durrett, Prof. Qiang Liu, Prof. Cho-jui Hsieh & Prof. Sewon Min

Indian Institute of Technology Bombay *2015 - 2019*
B.Tech (Honours) in Computer Science and Engineering
Advisor: Prof. Shivaram Kalyanakrishnan

Work Experience

Mirendil *Member of Technical Staff, June 2026 - present*

Google Research *Student Researcher, Fall 2024*
Advisor: Dr. Felix Yu
Worked on developing BlockRank, a specialized long-context architecture for scalable LLM-based Ranking and RAG

Google Deepmind *Student Researcher, Summer 2024*
Advisor: Dr. Prateek Jain
Worked on developing *learned* super vocabulary tokenizers for large language models and establishing their scaling laws

Google *Student Researcher, 2022-2023, Summer 2025*
Advisor: Prof. Inderjit Dhillon
Worked on algorithms for jointly learning search index and representations, and recently on LLM-guided hierarchical retrieval

Microsoft Research India *Research Fellow in Machine Learning and Optimization Group, 2019 - 2021*
Advisor: Dr. Manik Varma
Worked on algorithms of Extreme Classification leading to multiple top-tier publications and impact across Microsoft products

Publications

Preprints

* - equal contribution

- **Can Language Models Actually Retrieve In-Context? Drowning in Documents at Million Token Scale**
  Siddharth Gollapudi, Nilesh Gupta, Prasann Singhal and Sewon Min
  Under review, 2026
- **Compressing Many-Shots in In-Context Learning**
  Devvrit Khatri, Pranamya Kulkarni, Nilesh Gupta, Yerram Varun, Liqian Peng, Jay Yagnik, Praneeth Netrapalli, Cho-Jui Hsieh, Alec Go, Inderjit S Dhillon, Aditya Kusupati and Prateek Jain
  Under review, 2025

Conference Publications

- **HSEARCH-R1: Learning Compact Value Models for LLM-guided Hierarchical Search via RL**
  Nilesh Gupta, Nasib Ullah, Jinbin Zhang, Rohit Babbar and Inderjit S Dhillon
  *Empirical Methods in Natural Language Processing (**EMNLP**) Main Conference, 2026*
- **LLM-guided Hierarchical Search for End-to-end Reasoning Intensive Retrieval**
  Nilesh Gupta, Wei-Cheng Chang, Ngot Bui, Cho-Jui Hsieh and Inderjit Dhillon
  *Transaction of Machine Learning (**TMLR**), 2026*
- **LUCID: Attention with Preconditioned Representations**
  Sai-surya Duvvuri, Nirmal Patel, Nilesh Gupta, Inderjit Dhillon
  *International Conference on Machine Learning (**ICML**), 2026*

- **Autoregressive Ranking: Bridging the Gap Between Dual and Cross Encoders**
  Benjamin Rozonoyer, Chong You, Michael Boratko, Himanshu Jain, Nilesh Gupta, Srinadh Bhojanapalli, Andrew McCallum and Felix X. Yu
  *Workshop on FoGen @ **ICML**, 2026* **(Poster)**
- **Compute Allocation for Reasoning-Intensive Retrieval Agents**
  Sreeja Apparaju and Nilesh Gupta
  *Workshop on MemAgents @ **ICLR**, 2026* **(Poster)**
- **Scalable In-context Ranking with Generative Models**
  Nilesh Gupta, Chong You, Srinadh Bhojanapalli, Sanjiv Kumar, Inderjit Dhillon and Felix Yu
  *Neural Information Processing Systems (**NeurIPS**), 2025*
- **EHI: End-to-end Learning of Hierarchical Index for Efficient Dense Retrieval**
  Ramnath Kumar*, Anshul Mittal*, Nilesh Gupta, Aditya Kusupati, Inderjit Dhillon and Prateek Jain
  *Transaction of Machine Learning (**TMLR**), 2024*
- **Exploring Design Choices for Building Language-Specific LLMs**
  Atula Tejaswi*, Nilesh Gupta* and Eunsol Choi
  *Empirical Methods in Natural Language Processing **EMNLP**, 2024*
- **Dual-encoders for Extreme Multi-label Classification**
  Nilesh Gupta, Devvrit Khatri, Srinadh Bhojanapalli, Ankit S. Rawat, Prateek Jain and Inderjit Dhillon
  *International Conference on Learning Representations (**ICLR**), 2024*
- **Negative Mining-aware Mini-batching for Extreme Classification**
  Kunal Dahiya*, Nilesh Gupta*, Deepak Saini*, Akshay Soni, Yajun Wang, Kushal Dave, Jian Jiao, Gururaj, Prasenjit Dey, Amit Singh, Deepesh Hada, Vidit Jain, Bhawna Paliwal, Anshul Mittal, Sonu Mehta, Ramachandran Ramjee, Sumeet Agarwal, Purushottam Kar and Manik Varma
  *International Conference on Web Search and Data Mining (**WSDM**), 2023*
- **ELIAS: End-to-end Learning to Index and Search in Large Output Spaces**
  Nilesh Gupta, Patrick Chen, Hsiang-Fu Yu, Cho-jui Hsieh and Inderjit Dhillon
  *Neural Information Processing Systems (**NeurIPS**), 2022*
- **Generalized Zero-Shot Extreme Multi-Label Learning**
  Nilesh Gupta, Sakina Bohra, Yashoteja Prabhu, Saurabh Purohit and Manik Varma
  *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2021*
- **Extreme Regression for Dynamic Search Advertising**
  Yashoteja Prabhu, Aditya Kusupati, Nilesh Gupta and Manik Varma
  *International Conference on Web Search and Data Mining (**WSDM**), 2020* **(Long Oral)**
  *Workshop on eXtreme Classification: Theory and Applications @ **ICML**, 2020*

Selected Awards and Honors

- Awarded $50,000 Gemini Academic Program Award for research on LLM-guided retrieval
- Ranked 4th in ACM-ICPC Asia Regionals and 6th in ACM-ICPC India Online
- All India Rank 384 in JEE Advanced (IIT-JEE) 2015 among 150,000 candidates
- Awarded the prestigious KVPY Fellowship from Government of India
- Ranked 2nd in Regional Mathematics Olympiad (RMO) and among top 300 students in INMO

Teaching & Responsibilities

- *Graduate Teaching Assistantship* - Computer Science Department, UT Austin
  - Symbolic Programming - *Prof. Gordon S. Novak* — *Fall 2022*
  - Fundamentals of Machine Learning - *Prof. Inderjit Dhillon* — *Spring 2023, Spring 2025*
  - Principles of Machine Learning I - *Prof. Angela Beasley* — *Fall 2023, Spring 2024*
- *Undergraduate Teaching Assistantship* - Computer Science and Engineering, IIT Bombay
  - Computer Programming and Utilisation - *Prof. Ganesh Ramakrishnan* — *Autumn 2018*
  - Computer Programming and Utilisation - *Prof. Krishna S.* — *Autumn 2017*
  - Basic Calculus - *Prof. Amiya K. Pani* — *Autumn 2016*
- *MOOC Teaching Assistantship* - IITBombayX, edX
  - Data Structures and Algorithms - *Prof. Deepak B. Phatak* — *Spring & Autumn 2017*
- *Managing Extreme Classification Reading Group* - Microsoft Research India — *2020 - 2021*